LEVEL 5 · PART II

CI/CD

5

Goal: Build a fully automated pipeline from git push to verified production deployment with rollback.

RECOMMENDED TIME
2-3 weeks

TOPICS
12

PRACTICE / CAPSTONE
Pipeline: Git Push → GitHub → Lint → Tests → Build → Docker Build → Security Scan → Push Image → Deploy → Health Check → Notify.

Topics

#	Topic	What you will learn
1	Designing a CI/CD Pipeline End to End **p.933**	stages, fail-fast, lint, tests, build, scan, push, deploy, health check, notify
2	GitHub Actions for CI/CD **p.945**	reusable workflows, composite actions, matrix, caching, concurrency
3	Automated Testing in Pipelines **p.959**	unit/integration/e2e, test pyramid, service containers, flaky tests
4	Docker Image Builds and Registries in CI **p.971**	buildx, layer cache in CI, tagging strategy (sha/semver), pushing to GHCR/ECR
5	Secrets and Environment Variables **p.987**	per-env config, secret stores, OIDC, never in images, rotation
6	Build Artifacts **p.1003**	build once deploy many, artifact promotion, retention, provenance
7	Rollback **p.1019**	image rollback, versioned releases, rollback triggers, roll-forward
8	Zero-downtime Deployments **p.1033**	graceful shutdown, connection draining, readiness, health gates
9	Database Migrations in Pipelines **p.1049**	expand/contract, backwards compatible migrations, Flyway/Prisma/Alembic, locking pitfalls
10	Rolling Deployments **p.1067**	batch size, max surge/unavailable, trade-offs

#	Topic	What you will learn
11	Blue/Green Deployments **p.1081**	two environments, traffic switch, instant rollback, cost
12	Canary Deployments **p.1095**	progressive traffic shift, metrics-based promotion, Argo Rollouts/Flagger overview
★	Level capstone project **p.1111**	Pipeline: Git Push → GitHub → Lint → Tests → Build → Docker Build → Security Scan → Push Image → Deploy → Health Check → Notify.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

